Wednesday Reading Assessment: Unit 4, Magnetic Induction

Prof. Jordan C. Hanson

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1 Memory Bank

- $\epsilon = -N\Delta\phi_m/\Delta t$ ... Faraday's Law
- $\phi_m = \vec{B}\cdot\vec{A}$ ... Definition of magnetic flux

2 Faraday's Law

1. Suppose a uniform B-field of 0.01 T made an angle of 45 degrees with an wire loop with an area of 0.1 m^2. (a) Calculate ϕ_m, the magnetic flux. (b) Suppose the B-field drops to 0 T in 10 ms. What is $\Delta\phi_m/\Delta t$?

2. (a) An MRI technician moves his hand from a region of very low magnetic field strength into an MRI scanner's 2.00 T field with his fingers pointing in the direction of the field. Find the average emf induced in his wedding ring, given its diameter is 2.20 cm and assuming it takes 0.250 s to move it into the field. (b) Discuss whether this current would significantly change the temperature of the ring.

3. Consider Fig. 1. Suppose a constant current I is flowing through a rigid rod that hangs vertically from flexible leads, and suddenly a B-field appears within the rectangle formed by the rod and leads. **The B-field is into the page.** The vertical portion containing the battery and resistor is also rigid. The mass m and length L of the rod have been chosen such that $I = \lambda g/B$, where $\lambda = m/L$. (a) What is the magnitude of the Lorentz force, in terms of the variables given? (b) What is the magnitude of the weight force, in terms of the variables given? (c) Show that if these two forces are equal, $I = \lambda g/B$. If $\epsilon = 120$ V, $R = 240$ Ω, $m = 0.05$ kg, and $L = 10$ cm, what B-field strength is required to achieve balance?

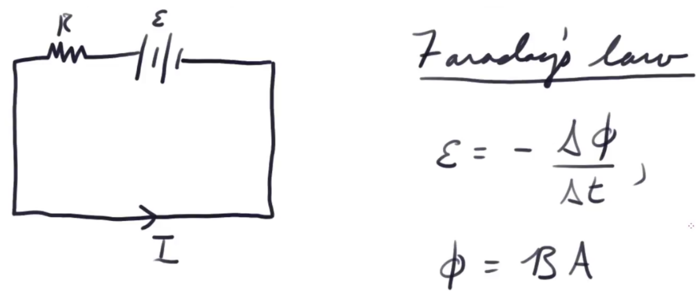

Figure 1: A current-carrying rod of length L hangs vertically in a B-field that turns on suddenly and *is in to the page.* The rod cannot change length, but the vertical leads are flexible and can change length.